Lab#4: Array Manipulations (Control Transfer Instructions).

Assignment #1: Write an assembly code to store the numbers (from 1 to 10) in the array x.

Solution:

```
01  ORG 100H
02          LEA SI,X
03          MOV AL,01H
04          MOV CX,000AH
05  L:      MOV [SI],AL
06          INC SI
07          INC AL
08          LOOP L
09  HLT
10  X DB 10 DUP(0)
```

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Assignment #2: Write an assembly code to find the maximum value in the array N.

N DW 100H, 50H, 250H, 30H

Solution:

```
01  ORG 100H
02              MOV SI,0
03              MOV CX,0004H
04              MOV BX,0000H
05  L:          CMP BX,N[SI]
06              JA ABOVE
07              MOV BX,N[SI]
08  ABOVE:      ADD SI,2
09              LOOP L
10              MOV MAX,BX
11  HLT
12  N DW 100H,50H,250H,30H
13  MAX DW ?
```

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Assignment #3: Write an assembly code to exchange the values in the two arrays

Solution:

```
01  ORG 100H
02              MOV SI,0
03              MOV AL,0
04              MOV BL,0
05              MOV CX,4
06  L:          MOV AL,X[SI]
07              MOV BL,Y[SI]
08              XCHG AL,BL
09              MOV X[SI],AL
10              MOV Y[SI],BL
11              INC SI
12              LOOP L
13  HLT
14
15  X DB 1,2,3,4
16  Y DB 5,6,7,8
```

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Assignment #4: Write an assembly code to exchange the values in the two arrays using stack (push and pop instructions)

Solution:

```
01  ORG 100H
02              MOV SI,0
03              MOV AX,0
04              MOV BX,0
05              MOV CX,4
06  L:          MOV AL,X[SI]
07              MOV BL,Y[SI]
08              PUSH AX
09              PUSH BX
10              POP AX
11              POP BX
12              MOV X[SI],AL
13              MOV Y[SI],BL
14               INC SI
15              LOOP L
16  HLT
17
18  X DB 1,2,3,4
19  Y DB 5,6,7,8
```

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Assignment #5: Write an assembly code to perform division without using DIV or IDIV instructions.

X = 25, Y = 3
AL = X / Y
AH = X % Y

Solution:

```
01  ORG 100H
02                  MOV AH, 25
03                  MOV BL, 3
04                  MOV CL, 0
05  LOOP:           CMP AH, BL
06                  JL FINISH
07                  SUB AH, BL
08                  ADD CL, 1
09                  JMP LOOP
10  FINISH:         MOV AL, CL
11
12  HLT
```

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